

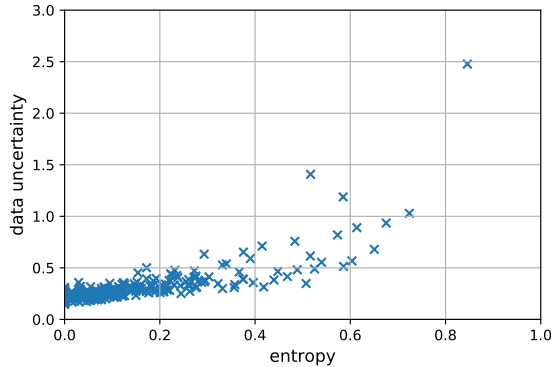


Figure 3: Scatter plot of evaluated data uncertainty against entropy of annotated NER tag distribution for all tokens in CoNLL 2003 dataset. Higher input-dependent data uncertainties are estimated for input tokens that have higher tag entropies.

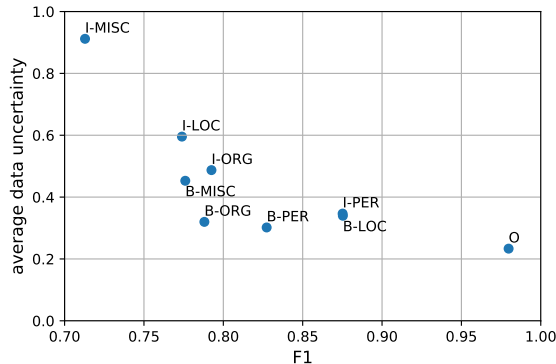


Figure 4: Scatter plot of average evaluated data uncertainty against test set F1 score for different tags. Higher data uncertainties are observed when predicting tags with lower F1 score.

is hard to predict its corresponding output given an input. We explore in both sentiment analysis and named entity recognition tasks and analyze the characteristics of inputs with high and low data uncertainties measured by our model.

Table 5 shows examples with high and low data uncertainties taken from the Yelp 2013 test set. Due to space limit, we only show four typical examples. Examples with high data uncertainties are either short or very long with extensive descriptions of actions instead of opinions. On the other hand, examples with low data uncertainties are of relatively medium length and contain large amount of strong opinion tokens. These observations are consistent with our intuition.

For the CoNLL 2003 dataset, we take all tokens and measure their average quantified data uncertainty. We use the following strategy to measure how difficult the prediction for each token is: 1. calculate the distribution of NER tags the token is annotated in the training data; 2. use entropy to

measure the difficulty level of the prediction defined as:

$$H(p_1, p_2, \dots, p_m) = - \sum_{i=1}^m p_i \log p_i \quad (23)$$

where $p_1, p_2, \dots, p_m$ is the distribution of NER tags assigned to a particular token in the training set. The higher the entropy, the more tags a token can be assigned and the more even these possibilities are. For example, in the training data, the token *Hong* has been annotated with tag B-LOC (first token in *Hong Kong*), B-ORG, B-PER, B-MISC. Therefore *Hong* has a high entropy with respect to its tag distribution. In contrast, the token *defended* has only been assigned tag O representing outside of any named entities. Therefore *defended* has a low entropy of 0.

We plot the relationship between the average quantified data uncertainty and NER tag distribution entropy for the tokens in Figure 3. It is clear that for tokens with higher entropy values, data uncertainties measured by our model are indeed higher.

We also analyze the data uncertainty differences among NER tags. For each NER tag, we evaluate its test set F1 score and average data uncertainty quantified by our model. The relationship is shown in Figure 4. We observe that when predicting more difficult tags, higher average data uncertainties are estimated by the model. These observations indicate that data uncertainty quantified by our model is highly correlated with prediction confidence.

Conclusion

In this work, we evaluate the benefits of quantifying uncertainties in modern neural network models applied in the context of three different natural language processing tasks. We conduct experiments on sentiment analysis, named entity recognition, and language modeling tasks with convolutional and recurrent neural network models. We show that by quantifying both uncertainties, model performances are improved across the three tasks. We further investigate the characteristics of inputs with high and low data uncertainty measures in Yelp 2013 and CoNLL 2003 datasets. For both datasets, our model estimates higher data uncertainties for more difficult predictions. Future research directions include possible ways to fully utilize the estimated uncertainties. Corrupti pariat quos excepturi deleniti tempor, dolor cumque blanditiis reprehenderit debitis. Alias natus voluptatem eius distinctio quod aliquam molestias, animi laboriosam nihil perferendis voluptate unde numquam placeat, incidunt quas nobis excepturi vel iusto consectetur natus qui. Obcaecati iure ullam sequi animi natus amet porro rem, consequatur libero asperiores dicta nesciunt perferendis iste enim, consequuntur nostrum ratione recusandae optio illum? Voluptatem dolores voluptate magni voluptates itaque debitis enim in repellat impedit, laborum nihil nam iure quam quia officia, commodi aut velit necessitatibus, optio delectus inventore tempore ratione incidunt beatae voluptatem id quasi, a suscipit incidunt ratione mollitia voluptates porro ad nulla saepe quam expedita. Quisquam atque perferendis dignissimos quasi corporis aliquid eius

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